

Banking Customer Analytics

1. Project Overview

This project analyzes banking customer data to understand customer relationships, financial behavior, and product utilization across various banking services. The goal is to uncover insights into customer segmentation, deposit and loan patterns, credit card usage, profitability, and risk exposure to support strategic business decisions and improve customer engagement.

2. Dataset Summary

- **Rows:** 3,001 Customer Records
- **Columns:** 25 - Banking Customer Attributes and Financial Metrics
- **Key Features:**
 - **Customer Demographics:** Age, Gender, Income, Nationality
 - **Banking Relationship:** Banking Relationship Type, Loyalty Classification
 - **Deposit Information:** Bank Deposits, Savings Account, Checking Account, Foreign Currency Account
 - **Loan Information:** Business Lending, Credit Card Balance
 - **Risk & Profitability:** Risk Weighting, Customer Profitability
 - **Financial Products:** Number of Bank Products Owned
 - **Customer Value Metrics:** Estimated Income, Properties Owned

3. Exploratory Data Analysis using Python

- **Environment Setup & Data Loading :**
 - First, we import the required Python libraries for data manipulation and visualization, then load the dataset “**Banking.csv**” using **pandas**.

- Initial Data Inspection (df.head()) :

```
[3]: df.head()
```

```
[3]:
```

	Client ID	Name	Age	Location ID	Joined Bank	Banking Contact	Nationality	Occupation	Fee Structure	Loyalty Classification	...	Bank Deposits	Checking Accounts	Saving Accounts	Foreign Currency Account	Business Lending
0	IND81288	Raymond Mills	24	34324	5/6/2019	Anthony Torres	American	Safety Technician IV	High	Jade	...	1485828.64	603617.88	607332.46	12249.96	113447.1
1	IND65833	Julia Spencer	23	42205	12/10/2001	Jonathan Hawkins	African	Software Consultant	High	Jade	...	641482.79	229521.37	344635.16	61162.31	200052.1
2	IND47499	Stephen Murray	27	7314	1/25/2010	Anthony Berry	European	Help Desk Operator	High	Gold	...	1033401.59	652674.69	203054.35	79071.78	54813.1
3	IND72498	Virginia Garza	40	34594	3/28/2019	Steve Diaz	American	Geologist II	Mid	Silver	...	1048157.49	1048157.49	234685.02	57513.65	114840.1
4	IND60181	Melissa Sanders	46	41269	7/20/2012	Shawn Long	American	Assistant Professor	Mid	Platinum	...	487782.53	446644.25	128351.45	30012.14	167441.1

5 rows × 25 columns

- Dataset Dimensions (df.shape) :

```
[4]: df.shape
```

```
[4]: (3000, 25)
```

- Dataset Overview & Info

```
[5]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 3000 entries, 0 to 2999
Data columns (total 25 columns):
#   Column                               Non-Null Count  Dtype
---  -
0   Client ID                           3000 non-null   str
1   Name                                3000 non-null   str
2   Age                                  3000 non-null   int64
3   Location ID                         3000 non-null   int64
4   Joined Bank                         3000 non-null   str
5   Banking Contact                     3000 non-null   str
6   Nationality                         3000 non-null   str
7   Occupation                          3000 non-null   str
8   Fee Structure                       3000 non-null   str
9   Loyalty Classification              3000 non-null   str
10  Estimated Income                    3000 non-null   float64
11  Superannuation Savings              3000 non-null   float64
12  Amount of Credit Cards              3000 non-null   int64
13  Credit Card Balance                 3000 non-null   float64
14  Bank Loans                          3000 non-null   float64
15  Bank Deposits                       3000 non-null   float64
16  Checking Accounts                   3000 non-null   float64
17  Saving Accounts                     3000 non-null   float64
18  Foreign Currency Account            3000 non-null   float64
19  Business Lending                    3000 non-null   float64
20  Properties Owned                    3000 non-null   int64
21  Risk Weighting                      3000 non-null   int64
22  BRId                                3000 non-null   int64
23  GenderId                            3000 non-null   int64
24  IAIId                               3000 non-null   int64
dtypes: float64(9), int64(8), str(8)
memory usage: 812.2 KB
```

- Generate Descriptive Statistics for the Dataframe

- Check for Missing Value

```
df.isnull().sum()
```

- Feature Preprocessing & Engineering :

- Standardizing column names by converting them to lowercase and replacing spaces with underscores.
- Check the updated column list.
- **Binning Income into Bands**
- Creating an **income_band** categorical feature based on **estimated_income**

- Categorical Feature Distributions :

- Examine the Distribution of Unique Categories in Categorical Columns
- **Output :-**
- **brid:** 3: 1352, 1: 660, 2: 495, 4: 493
- **genderid:** 2: 1512, 1: 1488
- **amount_of_credit_cards:** 1: 1922, 2: 765, 3: 313
- **nationality:** European 1309, Asian: 754, American: 507, Australian: 254, African: 176
- **fee_structure:** High: 1476, Mid: 962, Low: 562
- **loyalty_classification:** Jade: 1331, Silver: 767, Gold: 585, Platinum: 317
- **properties_owned:** 2: 777, 1: 776, 3: 742, 0: 705
- **risk_weighting:** 2:1222, 1: 836, 3: 460, 4: 322, 5: 160
- **income_band:** Med: 1517, Low:1027, High: 456

- Univariate Analysis :

- Visualizing distributions for **categorical features** across the **dataset**.

- Numerical Feature Distributions :

- Analyzing continuous variables such as **account balances**, **loan amounts**, and **age**.

- Bivariate & Multivariate Analysis :

- Correlation Analysis
- Examining the relationships between **continuous financial variables** using a **correlation heatmap**

4. Data Analysis using SQL :

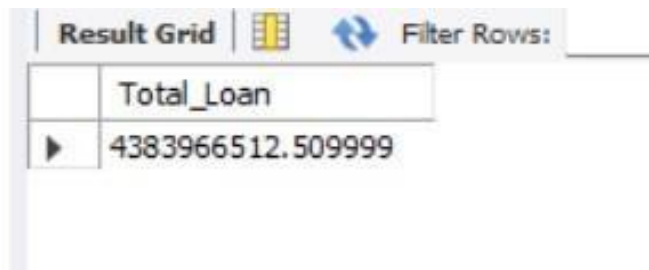
- Storing Data in a MySQL Database :
- To persist the cleaned and transformed dataset, we connect to a **MySQL server** using **sqlalchemy** and **pymysql** and create a dedicated database and table.
- Total Clients



A screenshot of a Jupyter Notebook's 'Result Grid' showing a single row with the column name 'Total_Clients' and the value '3000'. The interface includes a 'Filter Rows' button and a table icon.

	Total_Clients
▶	3000

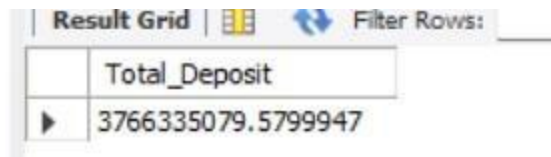
- Total Loans



A screenshot of a Jupyter Notebook's 'Result Grid' showing a single row with the column name 'Total_Loan' and the value '4383966512.509999'. The interface includes a 'Filter Rows' button and a table icon.

	Total_Loan
▶	4383966512.509999

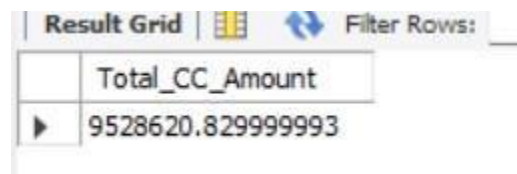
- Total Deposit



A screenshot of a Jupyter Notebook's 'Result Grid' showing a single row with the column name 'Total_Deposit' and the value '3766335079.5799947'. The interface includes a 'Filter Rows' button and a table icon.

	Total_Deposit
▶	3766335079.5799947

- Total Credit Card Amount



A screenshot of a Jupyter Notebook's 'Result Grid' showing a single row with the column name 'Total_CC_Amount' and the value '9528620.829999993'. The interface includes a 'Filter Rows' button and a table icon.

	Total_CC_Amount
▶	9528620.829999993

- Savings Account Amount

Result Grid		Filter Rows:
	Savings_Account	
▶	698725060.4500005	

- Bank Loan by Income Band

	Income_Band	Bank_Loan
▶	Med	942491549.6500002
	Low	448193267.85000044
	High	383473648.96

- Bank Loan by Nationality

	Nationality	Bank_Loan
▶	European	777615715.5199997
	Asian	436187379.57000005
	American	306027878.4300001
	Australian	154158054.45999995
	African	100169438.47999997

- Deposit by Income Band

	Income_Band	Bank_Deposit
▶	Med	1091330501.6799986
	Low	499351082.1400001
	High	423998997.9500001

- Deposit by Nationality

	Nationality	Bank_Deposit	Savings_Account	Checking_Account
▶	American	335300255.3400002	120163974.99999999	160282607.57999998
	African	121064820.11999999	38294856.15000002	57391085.75000001
	European	873965472.3000009	304999813.00000036	420250741.83000034
	Asian	515311922.45999974	178630361.23000002	244396869.02000022
	Australian	169038111.54999998	56636055.07000003	80957543.2

- Loyalty Classification vs Fees

	loyalty_classification	fee_structure	total_customers
▶	Gold	High	304
	Gold	Low	108
	Gold	Mid	173
	Jade	High	668
	Jade	Low	240
	Jade	Mid	423
	Platinum	High	140
	Platinum	Low	61
	Platinum	Mid	116
	Silver	High	364
	Silver	Low	153
	Silver	Mid	250

- Nationality vs Fees

	nationality	fee_structure	total_customers
▶	African	High	93
	African	Low	26
	African	Mid	57
	American	High	252
	American	Low	91
	American	Mid	164
	Asian	High	367
	Asian	Low	141
	Asian	Mid	246
	Australian	High	126
	Australian	Low	49
	Australian	Mid	79
	European	High	638
	European	Low	255
	European	Mid	416

- Drill Through Table Query

	nationality	occupation	loyalty_classification	estimated_income	bank_deposits	bank_loans	business_lending	credit_card_balance	risk_weighting
▶	American	Safety Technician IV	Jade	75384.77	1485828.64	776242.92	1134475.3	484.54	2
	African	Software Consultant	Jade	289834.31	641482.79	1270615.43	2000526.1	2256.88	3
	European	Help Desk Operator	Gold	169935.23	1033401.59	1052715.84	548137.58	4568.74	3
	American	Geologist II	Silver	356808.11	1048157.49	121195.06	1148402.29	4205	4
	American	Assistant Professor	Platinum	130711.68	487782.53	1048301.95	1674412.12	3779.49	3
	American	Help Desk Technician	Silver	118326.96	1307269.41	601902.5	1556031.06	5148.56	2
	Asian	Account Coordinator	Gold	57336.47	41200.18	208909.69	154111.62	959.9	1
	European	Automa Account Coordinator		65125.8	156983.13	1140704.8	1171456.68	4576.58	2
	Asian	Software Test Engineer II	Silver	87849.47	1242347.22	803444.46	464560.28	78.62	1
	European	Geologist III	Silver	65369.36	317246.67	60027.9	908583.94	4836.86	1
	European	Administrative Officer	Jade	74966.51	37979.77	214327.82	353119.93	493.32	1
	American	Database Administrator II	Jade	111449.06	749007.94	580468.51	294397.85	392.59	2
	African	Staff Scientist	Silver	115637.19	616330.93	815250.64	972455.46	5055.31	1
	American	Programmer I	Jade	124434.78	802862.97	984792.5	835554.83	5211.37	2
	Asian	Software Test Engineer I	Platinum	260795.34	116399.86	310876.62	1136169.3	2866.34	2

- Customers by Loyalty Classification

	loyalty_classification	total_customers
▶	Jade	1331
	Silver	767
	Gold	585
	Platinum	317

- Customers by Gender

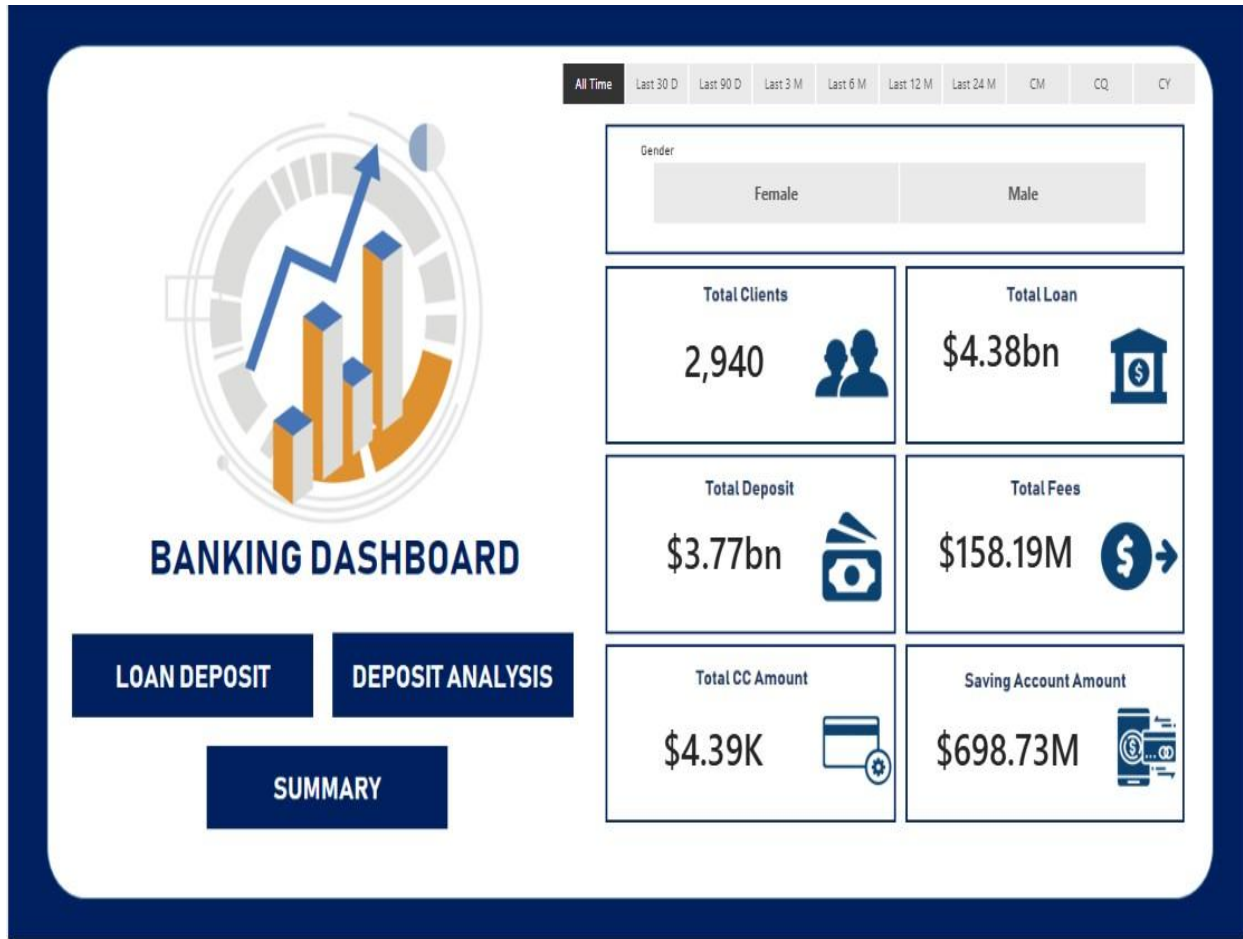
	genderid	total_customers
▶	2	1512
	1	1488

- Customers by Investment Advisor

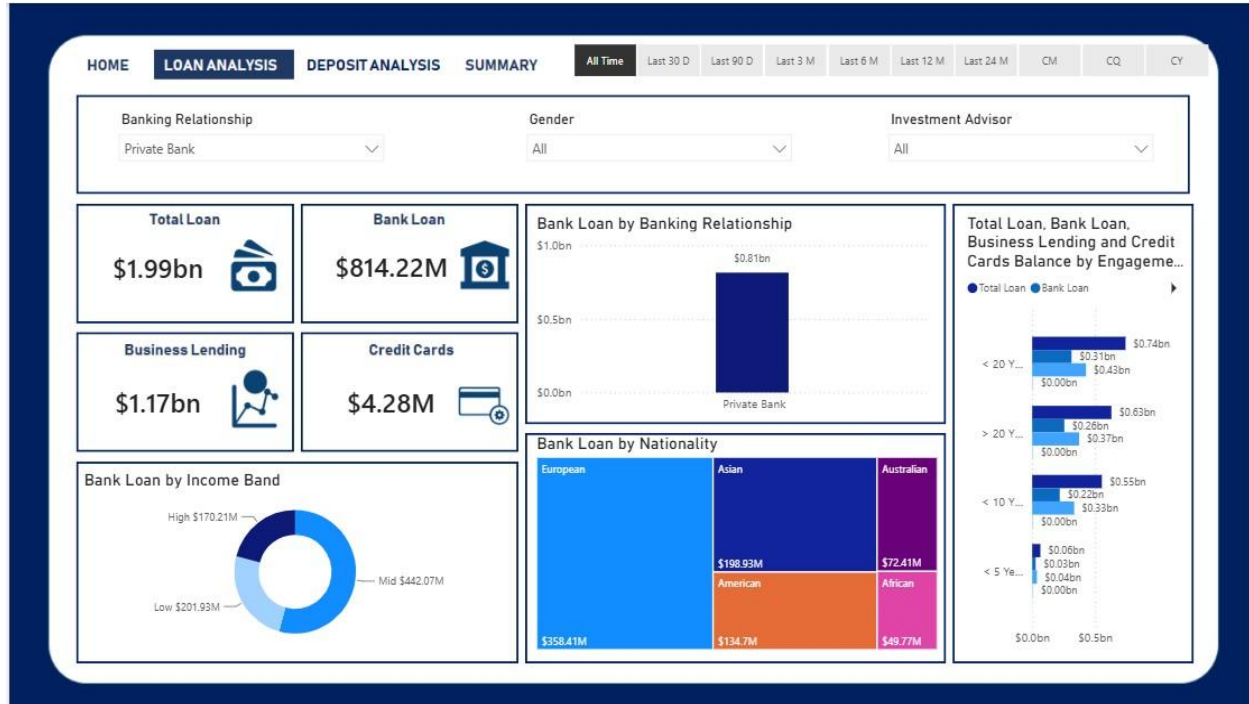
	iaid	total_customers
▶	1	177
	2	177
	3	177
	4	177
	8	177
	9	176
	10	176
	11	176
	12	176
	13	176
	14	176
	15	176
	5	89
	6	89
	7	89
	16	88
	17	88
	18	88
	19	88

5. Dashboard in Power BI :

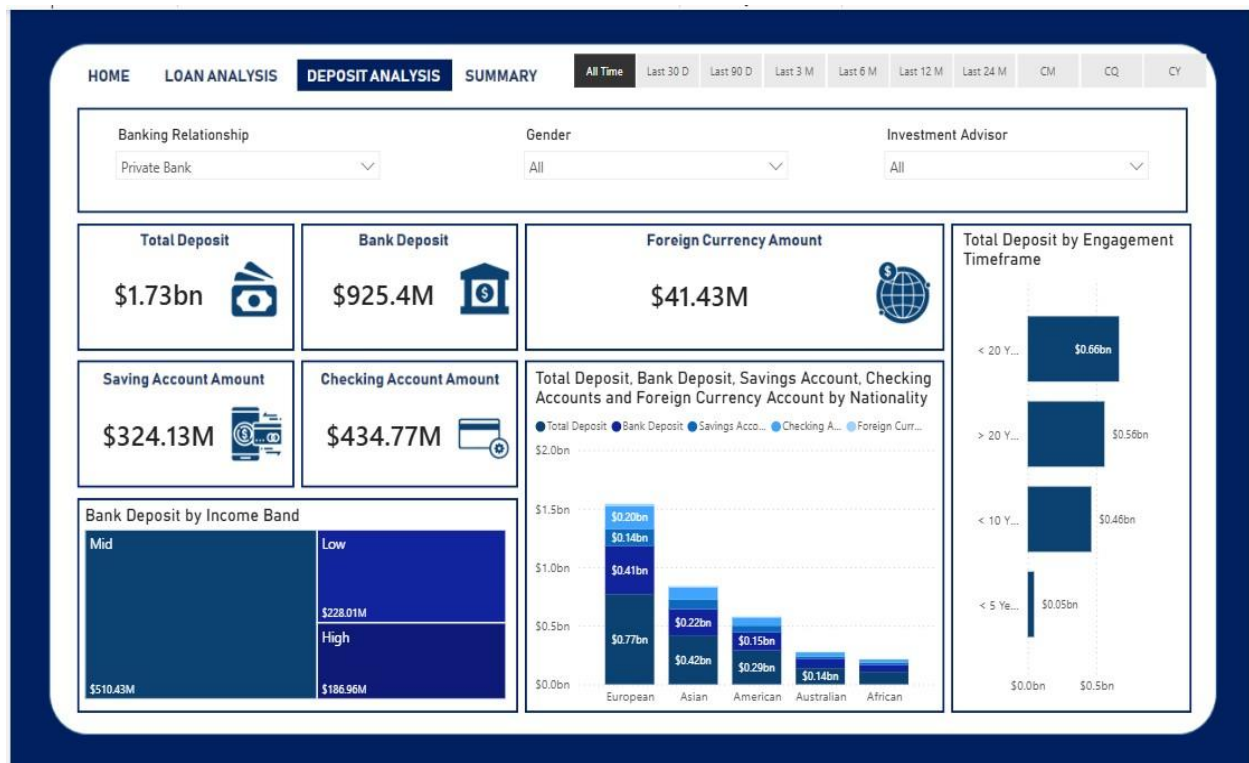
- Finally, we built an interactive dashboard in **Power BI** to present insights visually.
- Home Page



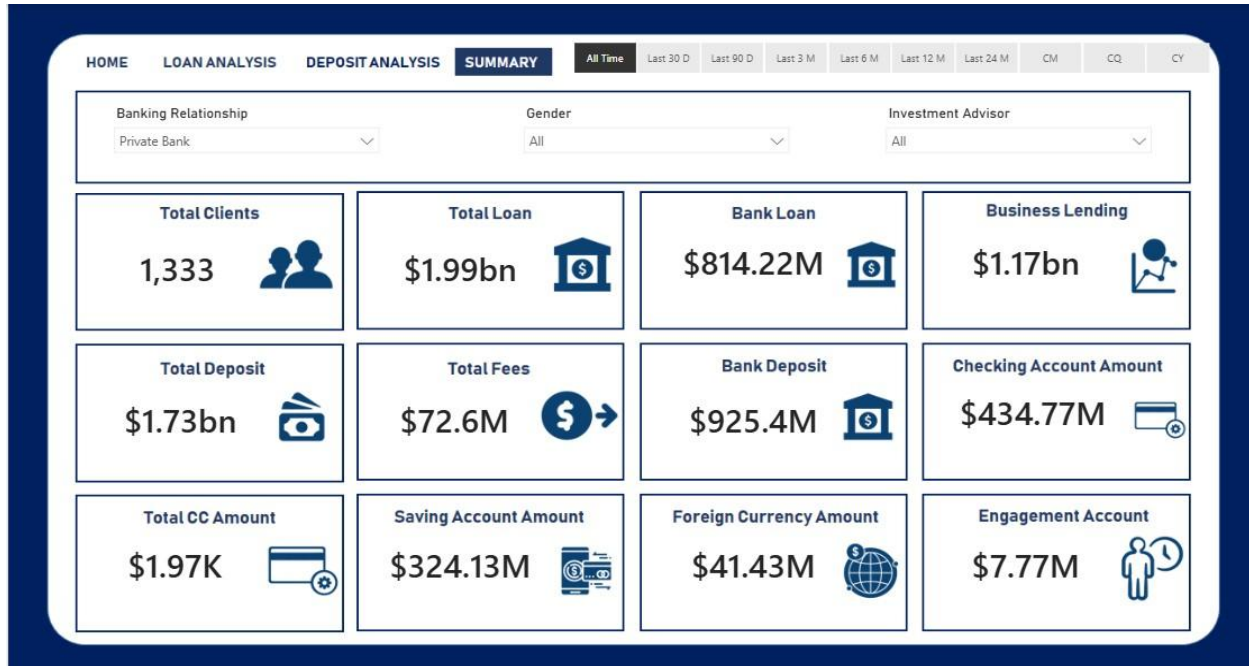
- Loan Analysis Page :



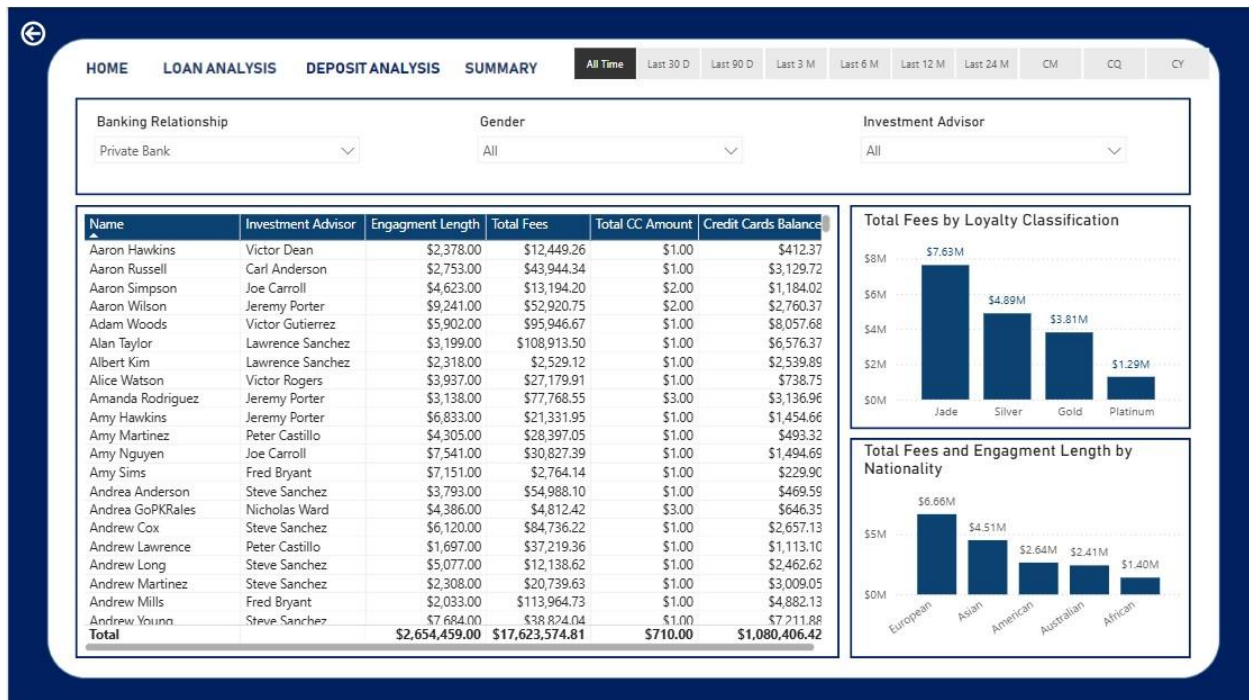
- Deposit Analysis Page :



- Summary Page :



- Drill Through Page :



6. Recommendations :

- **Increase Deposit Growth** – Promote savings and fixed-deposit products to strengthen the bank's deposit base.
- **Expand Loan Portfolio** – Target qualified customers with personalized loan offers to increase lending revenue.
- **Strengthen Customer Engagement** – Focus on customers with lower engagement levels through personalized banking services.
- **Promote Cross-Selling** – Encourage existing customers to adopt additional products such as credit cards, savings accounts, and loans.
- **Grow Business Lending** – Identify high-potential business clients and offer tailored financing solutions.
- **Increase Digital & Foreign Currency Services** – Promote foreign currency and digital banking products to diversify revenue streams.
- **Improve Customer Retention** – Develop loyalty programs for high-value customers to reduce churn and increase lifetime value.
- **Optimize Fee-Based Revenue** – Review banking service fees and introduce value-added services to enhance non-interest income.

